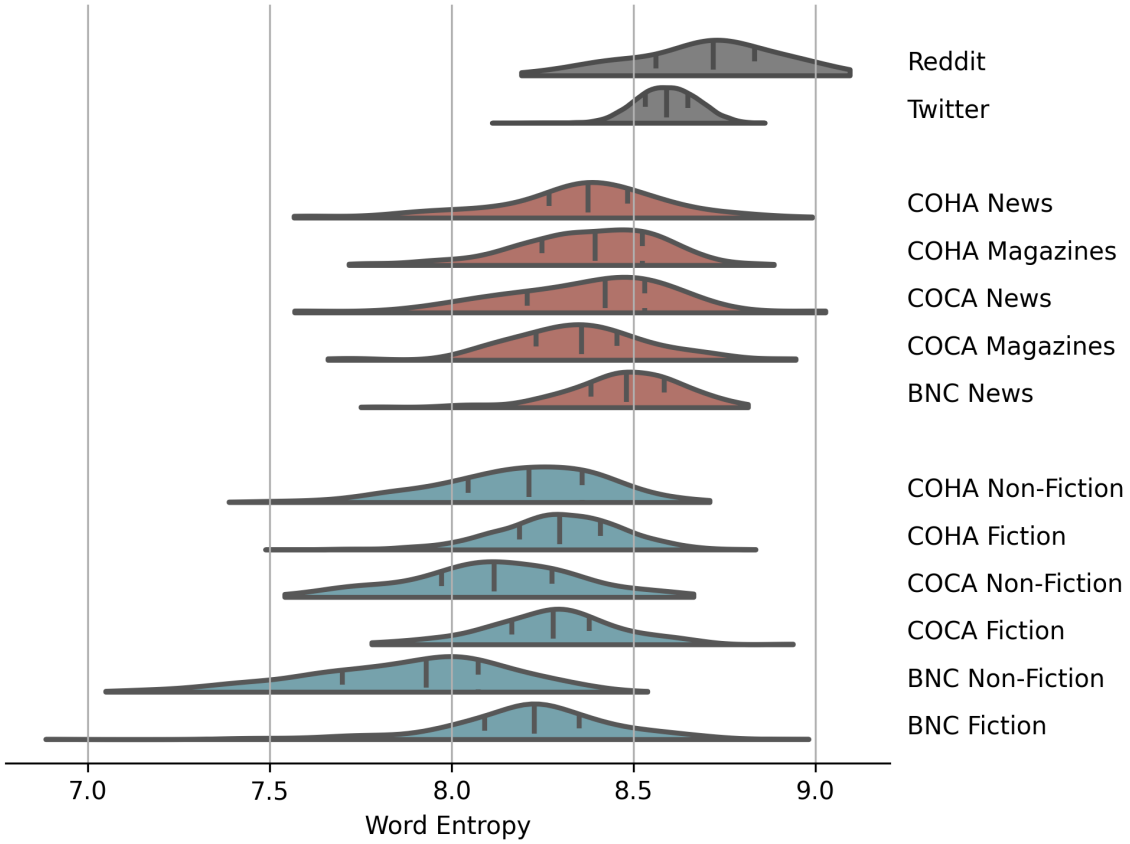

Figure 3: Word entropy of very short-form (social) media, short-form (news and magazines) and long-form (fiction and non-fiction) media. For each media category, distributions are kernel density estimates cut to the data range, with quartile positions shown. The COHA data was restricted to 2000-2007 to minimise the effect of historical changes.

models relate the consumption of information items with some utility gain to the forager. To bridge utility rates to lexical measures, we borrow the idea of information signal entropy from Shannon [35]: the entropy of a source of information is a function of the probability of seeing each symbol given the preceding symbols. For our purposes entropy can be thought of as a rate of information. If information foragers gain utility from information then, by definition, an increase in entropy, h, is associated with an increase in utility rate, r. This aligns with Zipf's principle of least effort [1, 4].

$$h \propto r\,. \tag{5}$$

Animal foragers modulate the selectivity of their diet in response to the environment, becoming more selective in times of abundance [36]. Why waste energy hunting difficult prey when there are plenty of easy calories around? Humans act in the same way when selecting information to consume [11, 37]. We have all experienced situations where we do not have access to the internet, for example on a plane or train journey, and we become less selective in what we read or watch.

This characterisation of attention corresponds to the prey choice model, which describes which types of prey are worth pursuing and consuming [36]. And this has been applied to information foraging before [11]. The derivation of the prey model followed here is exactly analogous to that found in the prey choice model in food foraging. Our contribution will come at the end of this section, where we extend the model to include media competition for attention.

Assume an information forager searches a media environment and encounters information of types, i, at Poisson rates λ_i. If consumed, information provides a benefit u_i in a handling time t_i, during which time the forager is not searching. Alternatively, the forager can choose to ignore information of a certain type and keep searching. The forager's choices to consume or ignore information determine the expected total time spent searching, T_s, and handling, T_h, information, as well as the total utility gain, U. Given

these constraints, the forager aims to optimise the expected overall rate of utility of foraging given by

$$R_{media} = \frac{U}{T_s + T_h} \,. \tag{6}$$

Here *media* describes the forager's local environment, such as a media platform. Media platforms are analogous to foraging patches in optimal foraging theory. The forager's choices of which information types to consume can be described as an information diet, D. The total expected utility is $U = \sum_D \lambda_i u_i T_s$. Similarly the total expected handling time is $T_h = \sum_D \lambda_i t_i T_s$. Substituting in and cancelling T_s, we can write the expected utility rate given a diet

$$R_{media} = \frac{\sum_D \lambda_i u_i}{1 + \sum_D \lambda_i t_i} \,. \tag{7}$$

Consuming an information item carries an expected opportunity cost of not spending that item's handling time looking for other items, equal to $t_i R_{media}$, and an expected utility gain of u_i. To maximise expected utility rate a forager should therefore consume the item if the item utility rate, $r_i = \frac{u_i}{t_i}$, is greater than the overall media platform utility rate, R_{media},

$$r_i \geq R_{media} \,. \tag{8}$$

This diet threshold condition is a familiar result from foraging theory [36, 38, 11]. To find the optimal diet, item types can be ranked in order of r_i and added to the diet one by one until this inequality fails [38]. See the Supplementary Information for a more thorough derivation.

We can now ask which information types a forager should include in their diet, D, to maximise their expected overall utility rate as a consequence of rising information prevalence, here λ_i. For items with $r_i < R_{media}$, increasing prevalence has no effect as these items are still not included in the diet. For items with $r_i \geq R_{media}$, increasing prevalence will mean more time spent handling these items and less time spent searching, so the overall media platform utility rate will increase,

$$\frac{\partial R_{media}}{\partial \lambda_i} \geq 0 \quad \forall i \,. \tag{9}$$

Combining this with the information diet criterion (Inequality 8), increasing information prevalence increases the information utility rate required for diet inclusion: foragers become more selective when prey (or information) is abundant, analogous to the prey model in optimal foraging theory [36].

We now extend traditional foraging theory to information co-evolution by asking how media producers respond to increasing selectivity among information foragers. By assuming there is some cost to media of producing more informative messages — a standard assumption underlying Zipf's principle of least effort [39, 1] — we conclude that an abundance of information creates an adaptive pressure that drives media producers to create information with a higher utility rate. A proxy for utility rate is information density, or word entropy. Figure 4 shows a simple simulation of this dynamic.

Competition Between Media Platforms Drives Differences Between Short- and Long-form Media

Information is distributed in media platforms (e.g., newspapers, magazines, books, Twitter, Reddit). The forager has to choose not only which information to consume within a media platform, but also which media platforms to visit. Analogous to the information choice model (Equation 8): an optimal information forager will visit a media platform if the expected media utility rate is greater than the background utility rate from foraging in the overall environment (see Supplementary Information for the full model),

$$R_{media} \geq R_{env} \,. \tag{10}$$

The utility rate of a media platform, R_{media}, is a summation over Poisson processes (Equation 7). To simplify this, let $\bar{u}_m$ be the average utility of information items consumed in the media platform, $\bar{t}_m$ the average time spent consuming information items, and λ_m the rate of encounter of any item in the diet. Equation 7 then becomes a variation of Holling's disc equation [40] (full derivation in Supplementary Information)

$$R_{media} = \frac{\lambda_m \bar{u}_m}{1 + \lambda_m \bar{t}_m} \,. \tag{11}$$

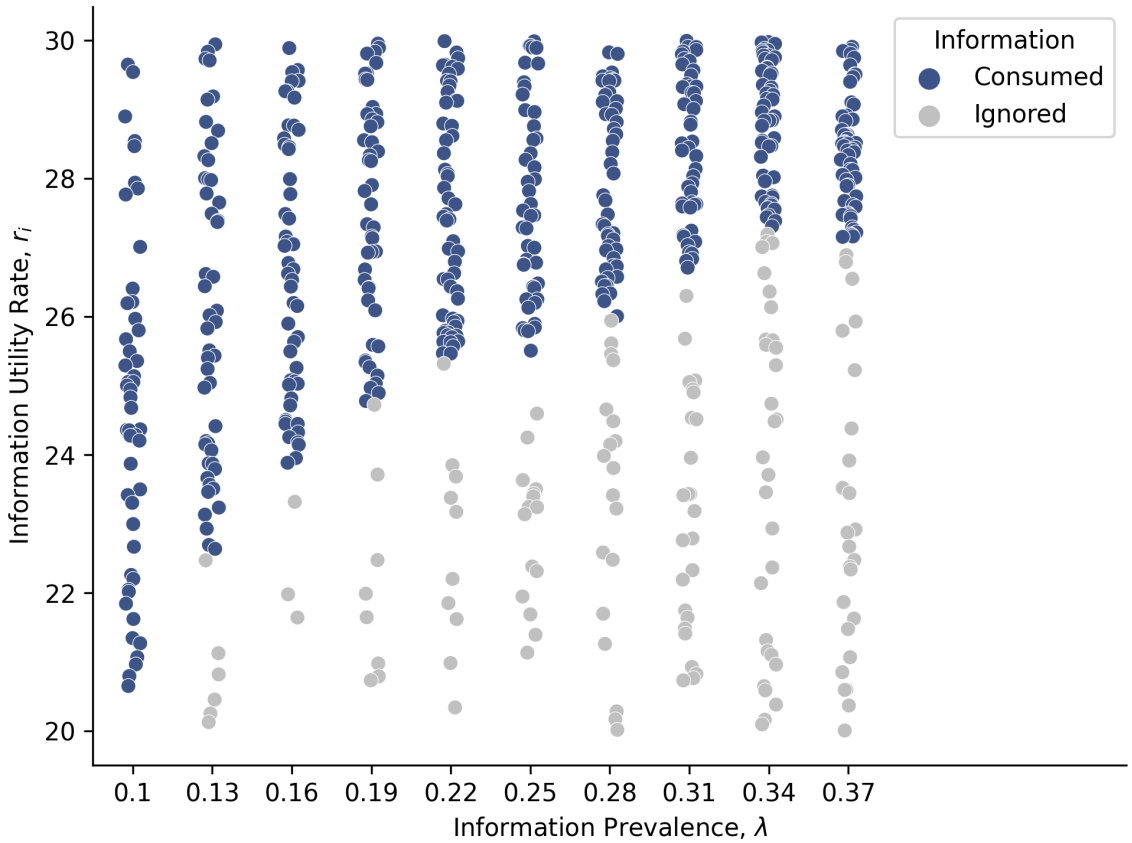

Figure 4: Simulation of information foraging in the attention economy. Information items are generated with random utility rates in quantities proportional to the information prevalence. Given the information environment, foragers only consume information items above a minimum information density (blue markers) in order to maximise their foraging rate. Information that is not consumed has less chance of survival (grey markers). Overall the surviving information types have higher utility rates at higher information prevalence.

This equation is visualised in Figure 5 **a**.

The criteria for inclusion in an information forager's diet is then

$$\frac{1}{\lambda_m \bar{u}_m} + \frac{1}{\bar{r}_m} \leq \frac{1}{R_{env}} \,. \tag{12}$$

The inclusion of a media platform in the information diet is therefore determined by three properties of the information items that it contains and which would be included in the forager's information diet: the average utility (i.e. size) of a item, $\bar{u}_m$; the average item utility rate, $\bar{r}_m$; and the prevalence of items within the media platform, λ_m.

Short-form media platforms such as news and magazines involve more time spent switching (and searching for) articles than long-form media platforms such as books. In order to reach the same overall media platform utility rate, R_{media}, short form media types need to have higher information utility rates (Figure 5 **c**). This creates a differential selective pressure on short- and long-form media producers. Given some R_{env}, the short-form media platform needs higher average information utility rates, $\bar{r}_m$, to be accepted in the forager's diet than the long-form media. The long-form media experiences a relaxed selective pressure on information utility rates because there is less time spent switching in these media platforms. This can describe the differences in the observed information utility rates in short- and long-form media as well as the trend towards increased information rates with increasing media prevalence.